

High-Level Project Timeline

Phase	Focus	Estimated Time	Key Deliverable
1	Data Acquisition & Infrastructure	Weeks 1-2	Automated data ingestion pipelines.
2	2D Mapping & ML Models	Weeks 3-5	Trained semantic segmentation model.
3	3D Simulation & UEHI Math	Weeks 6-8	The weighted 8-factor scoring algorithm.
4	Carbon Sink & Optimization	Weeks 9-10	Biomass-to-CO ₂ calculation engine.
5	Cloud Deployment & Dashboard	Weeks 11-12	Containerized REST API & Web UI.

Step-by-Step Execution Plan

Phase 1: Data Acquisition & Infrastructure (The Foundation)

What to do: Set up the environment and automate the fetching of your 8 geospatial datasets.

How to do it:

- **Define the AOI (Area of Interest):** Select one specific city ward to act as your initial Proof of Concept (PoC) to save compute costs.
- **Satellite Data:** Write Python scripts using the Google Earth Engine (GEE) API to pull Sentinel-2 (for multispectral bands like Near-Infrared) and Landsat data.
- **Ground Data:** Integrate the OpenStreetMap (OSM) API to pull road networks and building footprints.
- **Dynamic Data:** Set up API calls for real-time AQI (e.g., OpenWeather API) and download local census grids for the Population Exposure metric.

Phase 2: 2D Mapping & Feature Engineering (The Brain)

What to do: Build the Computer Vision model to identify trees, concrete, and soil.

How to do it:

- **Semantic Segmentation:** Use a pre-trained model like U-Net or Mask R-CNN. Fine-tune it on satellite imagery to classify pixels into three buckets: Canopy, Impervious Surface (Concrete/Asphalt), and Pervious Surface (Bare soil/Grass).
- **Index Calculation:** Use your multispectral data to calculate NDVI (Normalized Difference Vegetation Index) for every pixel to determine tree health.
- **Data Merging:** Overlay your segmentation map with the OSM building footprints to create a master 2D grid of the city block.

Phase 3: 3D Simulation & UEHI Math (The Core Logic)

What to do: Extrapolate the 2D map into 3D and calculate the 8-Factor UEHI Score.

How to do it:

- **3D Extrapolation:** Use Digital Surface Models (DSM) and Digital Terrain Models (DTM) from GEE to calculate building and canopy heights. Subtract DTM from DSM to get the actual volume of the structures and trees.
- **Scoring Algorithm:** Design the mathematical formula for the UEHI. You will need to normalize all 8 factors (NDVI, AQI, Temperature, Population, etc.) to a 0-1 scale. Assign statistical weights to each factor based on ecological importance (e.g., Canopy volume might hold a 20% weight, while AQI holds 15%) to generate the final 0-100 score.

Phase 4: Carbon Sink Planner & Optimization (The Value Proposition)

What to do: Calculate CO₂ offsets and determine optimal planting locations.

How to do it:

- **Biomass to CO₂ Conversion:** Apply standard allometric equations (often provided by forestry research) to convert the 3D canopy volume into Above-Ground Biomass (AGB), and then into tons of sequestered carbon.
- **The Optimizer:** Write a spatial optimization algorithm (like a Genetic Algorithm or simple linear programming). Set the objective to "Maximize Future Carbon & Health Score" with constraints like "Cannot plant on impervious surfaces" and "Must be within X meters of a water source."

Phase 5: Cloud Deployment & Dashboard (The Product)

What to do: Package the system so the SEER Bharat Foundation and CSR donors can actually use it.

How to do it:

- **Microservices:** Wrap your Python processing scripts and ML models into a REST API using FastAPI or Flask.
- **Containerization:** Use Docker to containerize the API so it runs consistently across any environment.
- **Cloud Hosting:** Deploy the Docker containers to AWS, Google Cloud, or Azure. Use a managed database (like PostgreSQL with PostGIS for spatial data) to store the processed

block scores.

- **Frontend:** Build a simple React or Vue.js dashboard with a map interface (using Mapbox or Leaflet) where users can click on a city block and see the UEHI score and carbon data.